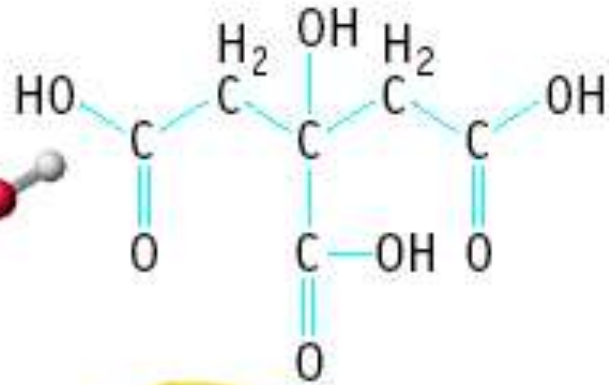
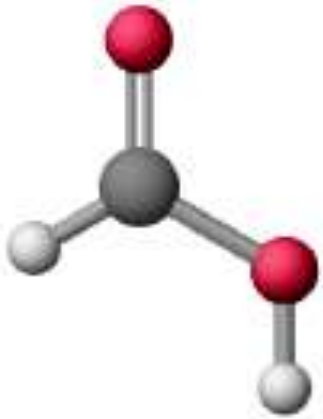
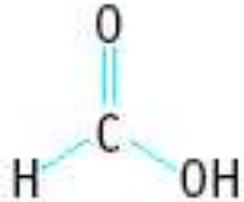
Acid and Bases





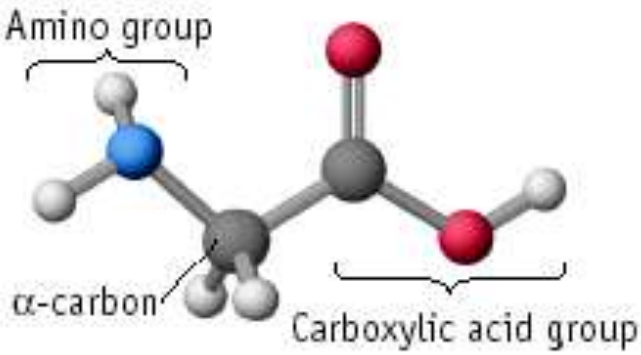


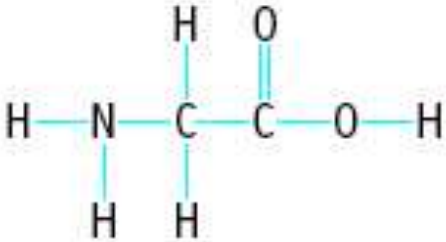
The tartness of lemons and oranges comes from the weak acid citric acid. The acid is found widely in nature and in many consumer products. *(Charles D. Winters)*

The sting of ants is due to the weak acid formic acid, HCO_2H. *(Gallo Images/@ CORBIS)*

Acid and Bases

Aspirin is a weak acid that has been used as an analgesic for over 100 years. *(Charles D. Winters)*





Glycine, $NH_2CH_2CO_2H$

Glycine is representative of the amino acids that are the basis of proteins. The $-CO_2H$ group is the acid portion of the molecule, and the $-NH_2$ group is the basic portion. *(Charles D. Winters)*

Some Properties of Acids

- ☑ Produce H^+ (as H_3O^+) ions in water (the hydronium ion is a hydrogen ion attached to a water molecule)
- ☑ Taste sour
- ☑ Corrode metals
- ☑ Electrolytes
- ☑ React with bases to form a salt and water
- ☑ pH is less than 7
- ☑ Turns blue litmus paper to red "Blue to Red A-CID"

Some Properties of Bases

- ☑ Produce OH^- ions in water
- ☑ Taste bitter, chalky
- ☑ Are electrolytes
- ☑ Feel soapy, slippery
- ☑ React with acids to form salts and water
- ☑ pH greater than 7
- ☑ Turns red litmus paper to blue "Basic Blue"

Some Common Bases

$NaOH$	sodium hydroxide	
KOH	potassium hydroxide	liquid soap
$Ba(OH)_2$	barium hydroxide	
$Mg(OH)_2$	magnesium hydroxide	
$Al(OH)_3$	aluminum hydroxide	

Acid/Base definitions

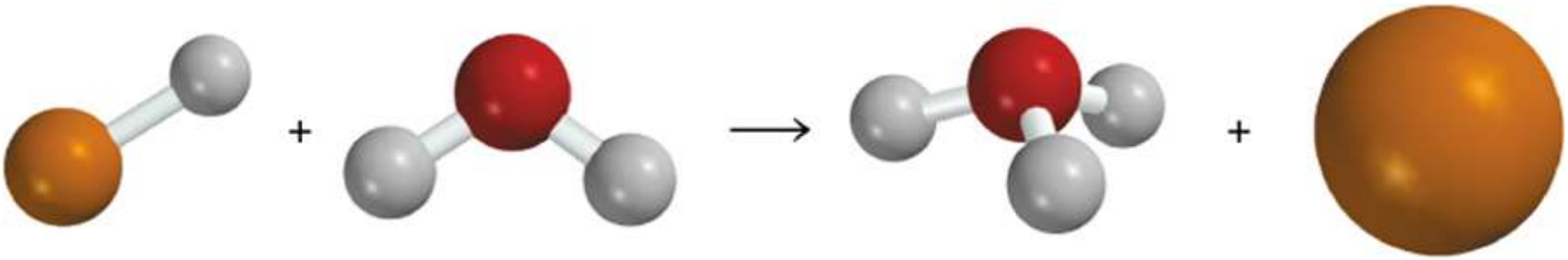
- Definition #1: Arrhenius (traditional)

Acids – produce H^+ ions (or hydronium ions H_3O^+)

Bases – produce OH^- ions

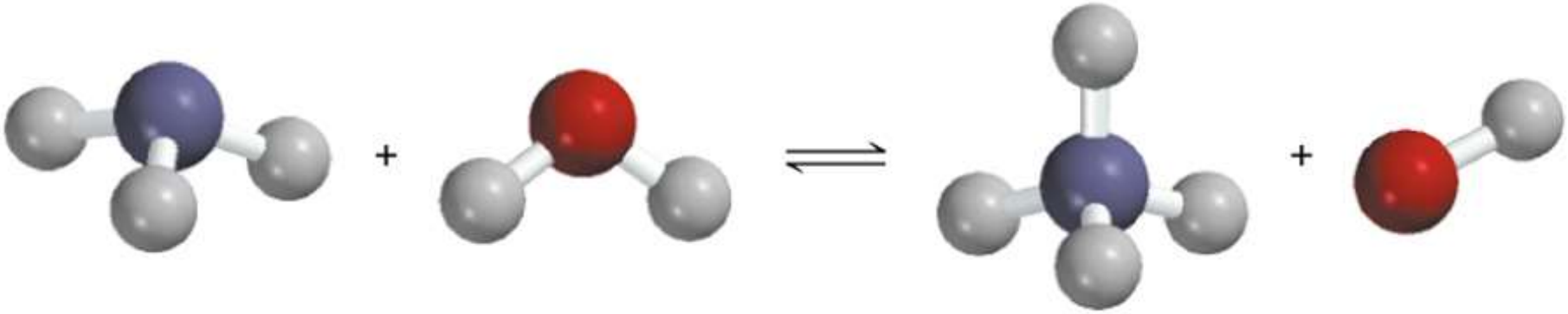
(problem: some bases don't have hydroxide ions

Arrhenius acid is a substance that produces H^+ (H_3O^+) in water



$$HCl + H_2O \longrightarrow H_3O^+ + Cl^-$$

Arrhenius base is a substance that produces OH^- in water



$$NH_3 + H_2O \rightleftharpoons NH_4^+ + OH^-$$

Acid/Base Definitions

- Definition #2: Brønsted – Lowry

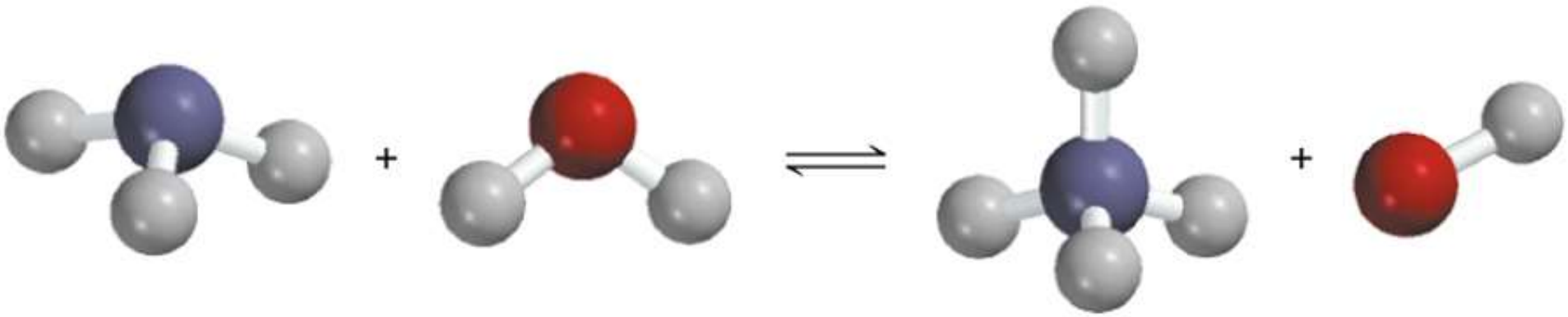
Acids – proton donor

Bases – proton acceptor

A “proton” is really just a hydrogen atom that has lost it’s electron!

A Brønsted-Lowry acid is a proton donor

A Brønsted-Lowry base is a proton acceptor



$$NH_3 + H_2O \rightleftharpoons NH_4^+ + OH^-$$

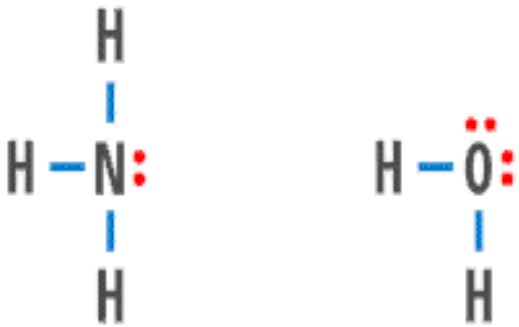
base acid conjugate acid conjugate base

ACID-BASE THEORIES

The Brønsted definition means NH_3 is a **BASE** in water — and water is itself an **ACID**

$$NH_3 + H_2O \rightleftharpoons NH_4^+ + OH^-$$

Base **Acid** **Acid** **Base**



Base Acid

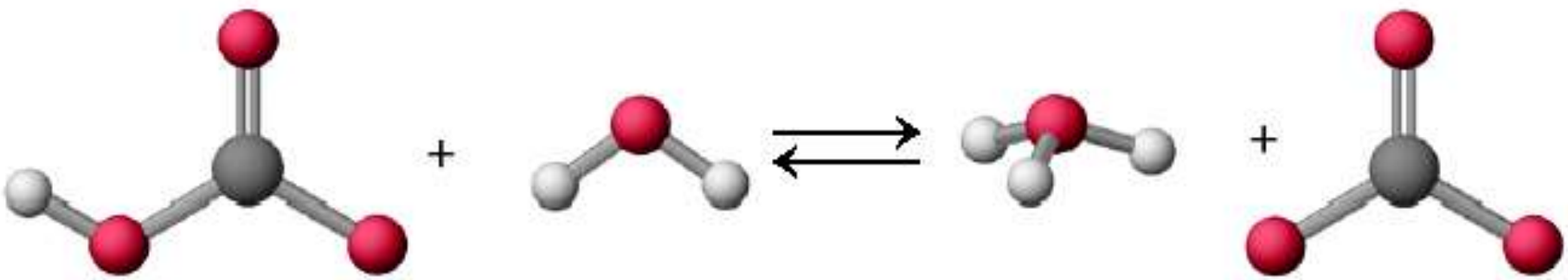
Conjugate Pairs

conjugate pair

conjugate pair

$HCO_3^-(aq)$ + $H_2O(\ell)$ $\rightleftharpoons$ $H_3O^+(aq)$ + $CO_3^{2-}(aq)$

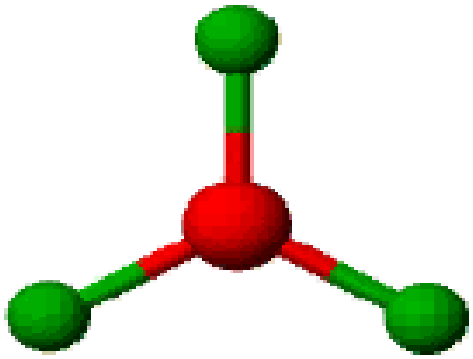
Acid *Base* *Acid* *Base*



Acids & Base Definitions

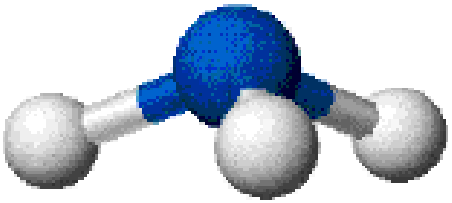
Definition #3 – Lewis

Lewis acid - a substance that accepts an electron pair



BF_3, the boron atom is surrounded by only three electron pairs.

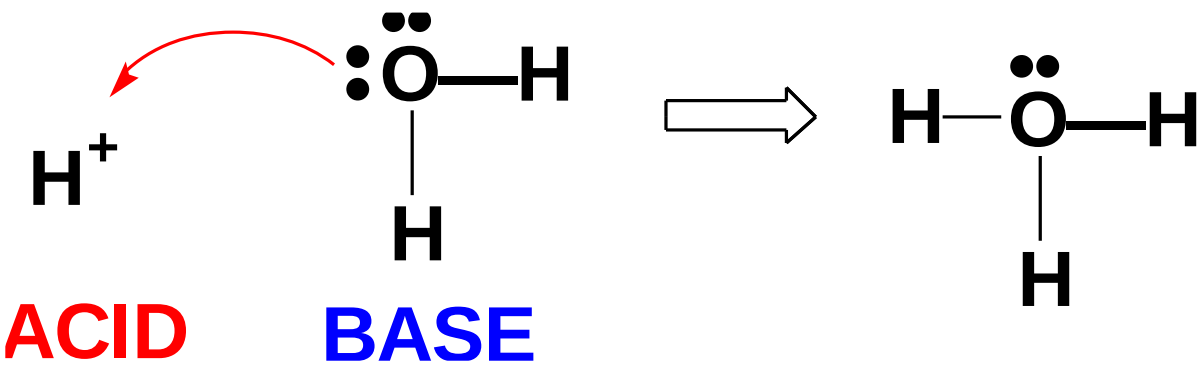
Lewis base - a substance that donates an electron pair



NH_3, the N atom has three bond pairs and one lone pair of electrons.

Lewis Acids & Bases

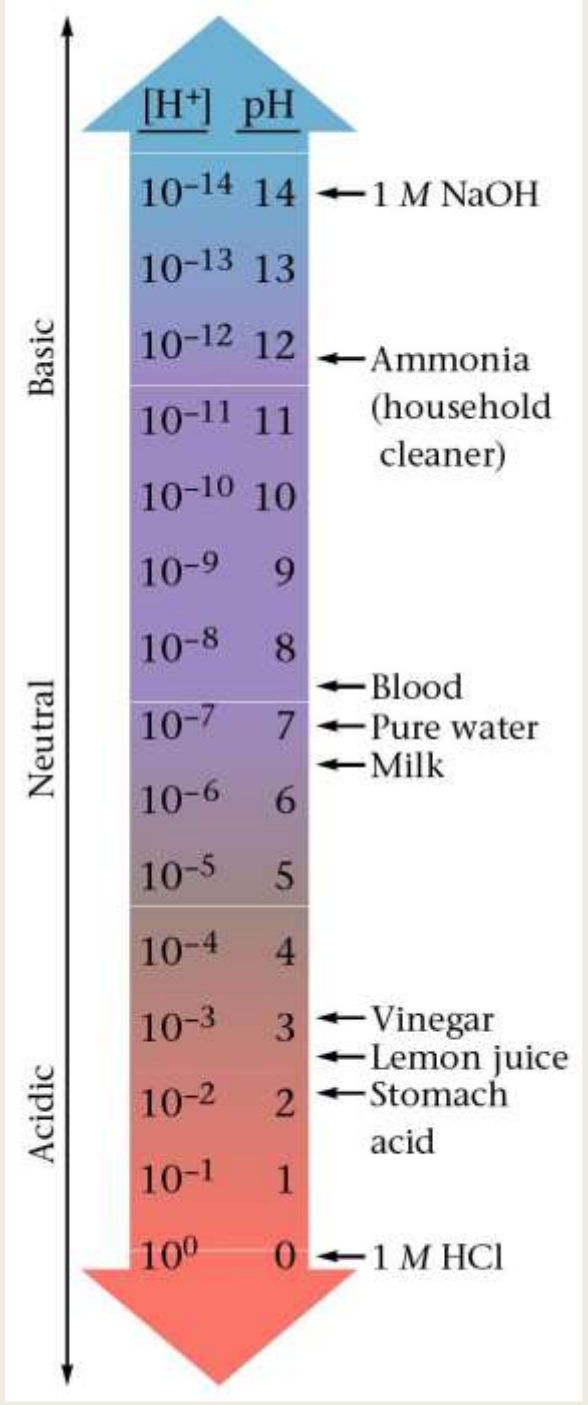
Formation of hydronium ion is also an excellent example.



- **Electron pair of the new O-H bond originates on the Lewis base.**

The pH scale is a way of expressing the strength of acids and bases. Instead of using very small numbers, we just use the NEGATIVE power of 10 on the Molarity of the H^+ (or OH^-) ion.

Under 7 = acid
7 = neutral
Over 7 = base



pH: The symbol "p" "means" the negative of the logarithm of "p" is often used very small or very large Values

- A pH meter

Calculating pH:

To calculating the pH o an aqueous solution you need to know the concentration of hydronium in mole/L (Molority). The pH in then calculating using the expression:

pH: $-\log[H^+]$

Calculating the pH

$$pH = -\log [H^+]$$

(Remember that the [ ] mean Molarity)

Example: If $[H^+] = 1 \times 10^{-10}$

$pH = -\log 1 \times 10^{-10}$

$pH = -(-10)$

$pH = 10$

Example: If $[H^+] = 1.8 \times 10^{-5}$

$pH = -\log 1.8 \times 10^{-5}$

$pH = -(-4.74)$

$pH = 4.74$

pH calculations – Solving for H+

- A solution has a pH of 8.5. What is the Molarity of hydrogen ions in the solution?

$$pH = -\log [H^+]$$

$$8.5 = -\log [H^+]$$

$$-8.5 = \log [H^+]$$

$$\text{Antilog } -8.5 = \text{antilog } (\log [H^+])$$

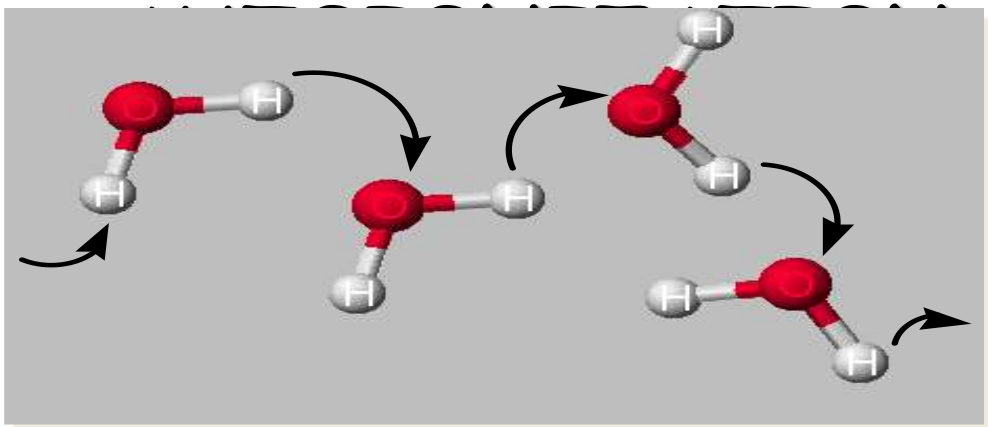
$$10^{-8.5} = [H^+]$$

$$3.16 \times 10^{-9} = [H^+]$$

More about water

H_2O can function as both an ACID and a BASE.

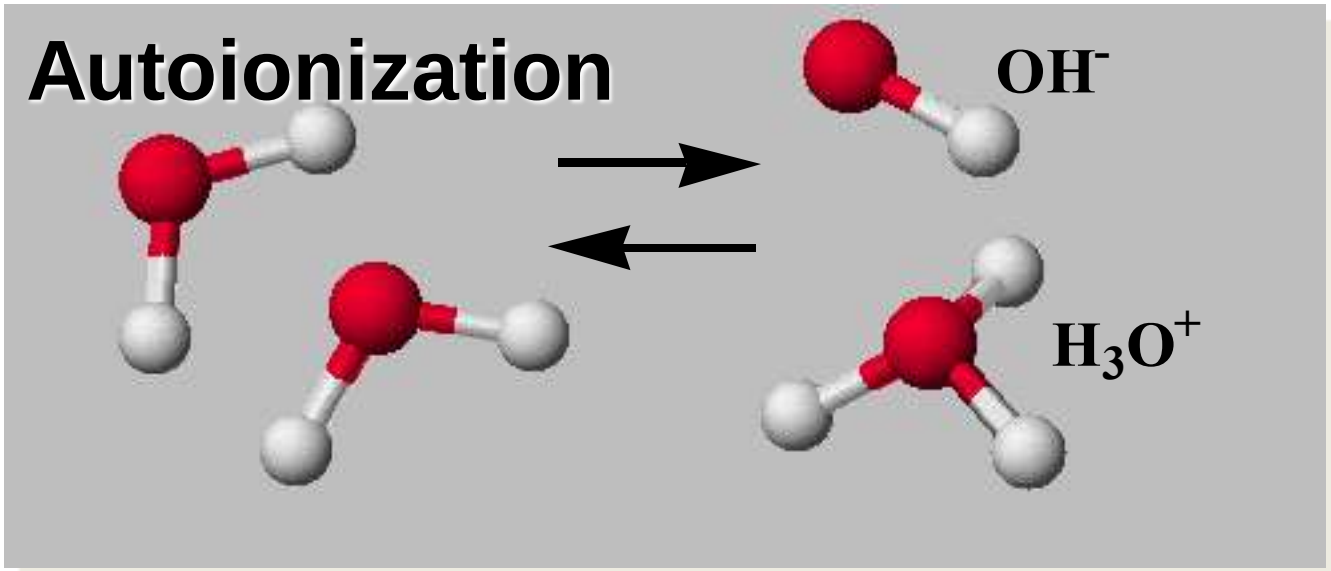
In pure water there can be



Equilibrium constant for water = K_w

$K_w = [H_3O^+][OH^-] = 1.00 \times 10^{-14}$ at 25 °C

More About Water



$K_w = [H_3O^+][OH^-] = 1.00 \times 10^{-14}$ at 25 °C

In a neutral solution $[H_3O^+] = [OH^-]$

so $[H_3O^+] = [OH^-] = 1.00 \times 10^{-7}$ M

pOH

- pOH looks at the perspective of a base

$$pOH = -\log [OH^-]$$

Since pH and pOH are on opposite ends,

$$pH + pOH = 14$$

Calculating poH:

To calculate the poH of a solution you need to know the concentration of the hydroxide ion in mole/L (molarity). The poH is then calculated using the expression:

poH= -log[oH-]

Relationship between pH and poH:The relationship between pH and poH the “ pH and poH of water solution at 25co are related by the following equation

$$pH + poH = 14$$

pH calculation in strong base solution

What is the pH of the 0.0010 M NaOH solution?

$[OH^-] = 0.0010$ (or 1.0×10^{-3} M)

$pOH = -\log 0.0010$

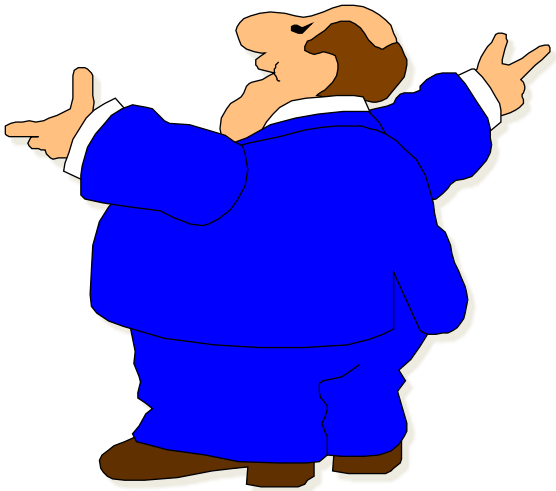
$pOH = 3$

$pH = 14 - 3 = 11$

OR $K_w = [H_3O^+][OH^-]$

$[H_3O^+] = 1.0 \times 10^{-11}$ M

$pH = -\log(1.0 \times 10^{-11}) = 11.00$



Example:

Find the pH of 0.0025M HCL solution

pH= $-\log[H^+]$=

$-\log[0.00025]=-(-2.60)=2.6$?

Calculating the Hydronium ion concentration from pH:

The hydronium concentration can be found from the pH by the reverse of the Mathematical operation employed to find the pH

$[H^+]=10^{-pH}$ or $[H^+]=\text{antilog}[-pH]$

Example:

What the hydronium concentration in a solution that has a pH of 8.34?

$8.34 = -\log[H^+]$

$-8.34 = \log[H^+]$

$[H^+] = 10^{-8.34} = 4.57 \times 10^{-9} M$

Calculating poH:

To calculate the poH of a solution you need to know the concentration of the hydroxide ion in mole/L (molarity). The poH is then calculated using the expression:

$$poH = -\log[oH^-]$$

Example:

What the poH of solution that has ahydroxide ion concentration of $4.82X10^{-5}M$?

$poH= -log[4.82X10^{-5}]=$

$-[4.32]=4.32$

Calculating the hydroxide ion concentration from poH:

The hydroxide ion concentration can be found from the poH by reverse mathematical operation employed to find the poH:

$[oH-]=10^{-poH}$ or $[oH^{-}]=antilog(-poH)$

Example:

What is hydroxide ion concentration solution that has poH of 5.70?

$5.70 = -\log[oH^-]$

$-5.70 = 10^{-5.70} = 2X10^{-6}M$

Calculator calculate $10^{-5.70}$ or "inverse" log(-5.70)

pH testing

- There are several ways to test pH
  - Blue litmus paper (red = acid)
  - Red litmus paper (blue = basic)
  - pH paper (multi-colored)
  - pH meter (7 is neutral, <7 acid, >7 base)
  - Universal indicator (multi-colored)
  - Indicators like phenolphthalein

HOME WORK

Find the pH of these:

1) A 0.15 M solution of Hydrochloric acid

2) A 3.00 X 10^{-7} M solution of Nitric acid

3)The pH of rainwater collected in a certain region of the northeastern United States on a particular day was 4.82. What is the H+ ion concentration of the rainwater?

4)The OH- ion concentration of a blood sample is 2.5 x 10-7 M. What is the pH of the blood?